

Granulite Facies Metamorphism, Two-Pyroxene Equilibria & Mafic Metabasites

Authoritative Reference: John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 25: Metamorphic Facies and Metamorphosed Mafic Rocks)

Comprehensive Chapter Masterclass: Metamorphic Facies & Mafic Rocks

The Eskola Facies System, Hornblende Dehydration Breakdown, ACF Chemographics & Two-Pyroxene Granulites

1. The Metamorphic Facies Concept & Mafic Protoliths

The concept of **metamorphic facies** was formulated by the Finnish petrologist **Pentti Eskola (1915, 1920, 1939)** to provide a rigorous, chemical-thermodynamic framework for correlating metamorphic rocks of widely differing bulk compositions. Eskola recognized that while pelites, carbonates, and basalts yield vastly different minerals under the same physical conditions, all rocks that equilibrated under identical temperature and pressure ranges belong to the same *metamorphic facies*.

"A metamorphic facies is a set of metamorphic mineral assemblages, repeatedly associated in time and space and showing a regular relationship between mineral composition and bulk chemical composition, such that they were formed under similar ranges of temperature and pressure... Because mafic rocks (basalts and gabbros) are chemically sensitive to P and T across almost the entire spectrum of crustal conditions, their mineral assemblages serve as the definitive standard for naming and defining metamorphic facies."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 25, pp. 568–571)

metamorphism. The mineral assemblages that develop in mafic rocks at low pressure do not differ significantly from those of medium to high pressure. Do we thus reject the albite-epidote hornfels and hornblende hornfels facies because the mineralogy is essentially the same as in the greenschist and amphibolite facies, respectively, so they cannot be defined on the basis of unique mineral assemblages? Or do we accept them as representing important low-pressure environments?

Texts appear to fall in about equal numbers on both sides of this argument. I have decided to include the hornfels facies types (although the IUGS-SCMR does not [Fettes and Desmons, 2007], and I have left the boundaries as dashed lines). Although there may not be a significant difference in the mafic mineral assemblages between low-pressure and

high-pressure environments, the boundaries will be discussed in section 5.1.

The “typical” continental geothermal gradient represents an average gradient for stable continental interiors. The local gradient can be either higher or lower, depending on the tectonic environment. See Spear (1993) for a discussion of various geotherm models.

As Yardley (1989) pointed out, it is convenient to consider metamorphic facies in four groups:

1. Facies of high pressure. The blueschist and eclogite facies are characterized by the development of low molar volume phases and assemblages under conditions of high pressure. The lower-temperature blueschist facies occurs in areas of unusually low T/P gradients, characteristically

FIGURE 2 Temperature–pressure diagram showing the generally accepted limits of the various facies used in this text. Boundaries are approximate and gradational. The “typical” or average continental geotherm is from Brown and Mussett (1993). The 30°C/km geothermal gradient is an example of an elevated orogenic geothermal gradient. The “forbidden zone” is bounded by the 5°C/km (150°C/GPa) gradient beyond which conditions are generally considered unrealized in nature (e.g., Liou et al., 2000).

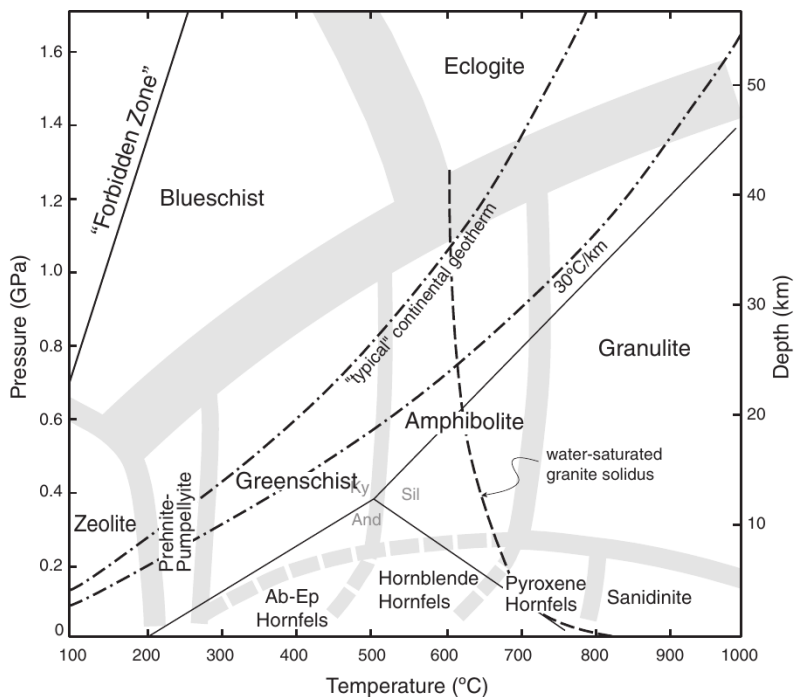


Figure 25-2 from John D. Winter (p. 570): Temperature–pressure diagram showing the definitive metamorphic facies defined for mafic protoliths (after Turner, 1981; Spear, 1993; Bucher and Frey, 1994). The granulite facies occupies the high-temperature region (> 700°C–800°C) across moderate to deep crustal pressures (0.3 to 1.2 GPa).

TABLE 1 Definitive Mineral Assemblages of Metamorphic Facies

Facies	Definitive Mineral Assemblage in Mafic Rocks
Zeolite	zeolites: especially laumontite, wairakite, analcime (in place of other Ca-Al silicates such as prehnite, pumpellyite, and epidote).
Prehnite–pumpellyite	prehnite + pumpellyite (+ chlorite + albite)
Greenschist	chlorite + albite + epidote (or zoisite) + actinolite ± quartz
Amphibolite	hornblende + plagioclase (oligoclase, andesine) ± garnet
Granulite	orthopyroxene + clinopyroxene + plagioclase ± garnet
Blueschist	glaucophanite + lawsonite or epidote/zoisite (± albite ± chlorite ± garnet)
Eclogite	pyrope garnet + omphacitic pyroxene (± kyanite ± quartz). No plagioclase.
Contact facies	mineral assemblages in mafic rocks of the facies of contact metamorphism do not differ substantially from those of the corresponding regional facies at higher pressure.

After Spear (1993), Fettes and Desmons (2007).

developed in subduction zones. Although the term *glaucophanite schist facies* has precedence over *blueschist facies*, having been used earlier, the latter term is more common. This is due to the blue-gray color of most mafic blueschists, imparted by sodic amphiboles. Eclogites occur in subduction zones, and, because they are also stable under normal geothermal conditions, they also develop wherever mafic rocks occur in the deep crust or mantle (stranded crustal magma chambers and dikes, or sub-crustal magmatic underplates). We will explore some recent discoveries of

prehnite–pumpellyite facies are thus treated separately. These facies are best developed where the protolith is immature and susceptible to metamorphism and where there are a high geothermal gradient and abundant hydrous fluids. As a result, they are most common in areas of burial or hydrothermal metamorphism affecting immature volcanics. Due to the poorly developed mineral assemblages and the complex dependency of these two facies types on mineral compositions, many petrologists have given up trying to distinguish between them and have taken to using the term **sub-green-**

Table 25-1 from John D. Winter (p. 571): Definitive mineral assemblages of metamorphic facies in mafic rocks. Note explicitly for the **Granulite Facies**: *Orthopyroxene + Clinopyroxene + Plagioclase ± Garnet ± Quartz*.

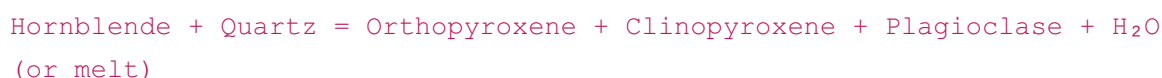
2. The Amphibolite-to-Granulite Transition: Hornblende Breakdown

In mafic protoliths (metabasites), the lower-grade **amphibolite facies** is characterized by the essential hydrous mineral assemblage **hornblende + plagioclase (oligoclase–andesine) ± quartz ± garnet**.

As temperature rises beyond approximately **700°C to 750°C** in the deep continental crust, hydrous amphibole becomes thermodynamically unstable and undergoes prograde **dehydration-devolatilization** (and dehydration-melting if quartz and plagioclase are present). The breakdown of hornblende produces anhydrous pyroxenes:

Dehydration Breakdown of Hornblende (Low to Moderate Pressure Granulites)

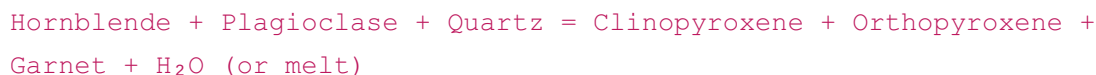
At moderate crustal pressures (3–8 kbar), hornblende reacts with quartz to generate orthopyroxene, clinopyroxene, and calcic plagioclase:



This reaction generates the classic **two-pyroxene granulite** (orthopyroxene + clinopyroxene + plagioclase), marking the definitive boundary of the granulite facies.

Dehydration Breakdown of Hornblende (High Pressure Granulites)

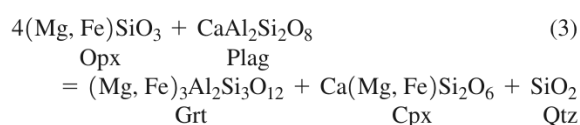
At higher lithostatic pressures (> 8–12 kbar in thickened orogenic roots), garnet joins the product assemblage:



High-pressure mafic granulites are thus characterized by the 5-phase anhydrous paragenesis:

Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz.

reaches the point that plagioclase is no longer stable. Metabasites in low pressure granulites lack garnet, whereas very-high-pressure granulites may lose orthopyroxene by reaction with plagioclase such as:



This reaction involves only solid phases and has a large ΔV associated with it. It is therefore pressure sensitive (low slope on a P - T phase diagram, as indicated by the Clapeyron equation). The reactant side is thus stable at low pressure, and the product side at high pressure. The reaction occurs at pressures higher than those of ~ 1.3 GPa at 800°C .

It is a long-held tenet that granulite facies terranes represent the deeply buried and dehydrated roots of the deep continental crust. Indeed, most exposed granulite facies rocks are found in the deeply eroded areas of the Precambrian continental shields. In his extensive review, however, Harley (1989) demonstrated that granulites are highly diverse, and Bohlen and Metzger (1989) showed that many have mid-crustal, not deep-crustal, origins. Touret (1971, 1985) showed that the fluid inclusions in granulite facies rocks of southern Norway are CO_2 rich, whereas those in the associated amphibolite facies rocks are more H_2O rich. This may imply that dehydration of at least some granulite facies terranes may be due to the infiltration of CO_2 replacing H_2O , rather than the

ing pressure. Similarly, the hornblende hornfels facies correlates with the amphibolite facies, and the pyroxene hornfels facies correlates with the granulite facies (although olivine is stable with plagioclase in the pyroxene hornfels facies but not in the granulite facies). The facies of contact metamorphism can be distinguished from those of medium-pressure regional metamorphism much better in metapelites (in which andalusite and cordierite develop).

Metamorphic Facies	Metamorphic Grade $\longrightarrow$						
	Greenschist	Transitional States	Amphibolite		Granulite		
Albite		---					
Plagioclase		Oligoclase				Andesine	
Epidote		---					
Actinolite		---					
Hornblende		---					
Augite							
Orthopyroxene							
Chlorite		---					
Garnet		---					
Biotite		---					
Quartz		---					
Phengite		---					
Cummingtonite		---					
Zone for associated metapelites	Chlorite Zone	Biotite Zone	Garnet Zone	Staurolite and Kyanite Zones	Sillimanite-Muscovite Zone	K-feldspar-Sillimanite Zone	Cordierite-Garnet Zone

FIGURE 10 Typical mineral changes that take place in metabasic rocks during progressive metamorphism in the medium P/T facies series. The approximate location of the pelitic zones of Barrovian metamorphism are included for comparison.

Figure 25-10 from John D. Winter (p. 577): Typical mineralogical changes occurring in metabasites across progressive metamorphic facies. Notice the complete termination of hornblende and epidote, replaced by the simultaneous incoming of orthopyroxene (Opx), clinopyroxene (Cpx), and garnet (Grt) in the granulite facies.

3. ACF Chemographic Topology of Granulite Facies Mafic Rocks

In Eskola's ACF diagram (where $A = \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3 - (\text{Na}_2\text{O} + \text{K}_2\text{O})$, $C = \text{CaO}$, $F = \text{FeO} + \text{MgO} + \text{MnO}$), projected from quartz and fluid, the granulite facies metabasites occupy the central three-phase and four-phase sub-triangles connecting Plagioclase (Anorthite), Diopside (Cpx), Enstatite-Ferrosilite (Opx), and Almandine-Pyropite (Grt).

Figure 1, and recommended by the IUGS-SCMR (Fettes and Desmons, 2007)—may be useful in higher-*P/T* terranes. In low *P/T* terranes and in contact aureoles, the plagioclase transition occurs first, resulting in a transitional zone with coexisting actinolite + calcic plagioclase (as calcic as labradorite). The transition to amphibolite facies usually occurs in the garnet zone of associated pelitic rocks (occasionally in the upper biotite zone).

Metabasites in the **amphibolite facies** are characterized by the assemblage hornblende + plagioclase ($An_{>17}$), with lesser amounts of garnet, clinopyroxene, quartz, and/or biotite (perhaps with some chlorite or epidote in the lower-grade portions of the facies). The amphibolite facies corresponds generally to the upper garnet, staurolite, kyanite, and lower sillimanite zones in associated pelitic rocks. Figure 8 is an *ACF* diagram illustrating the typical assemblages. Because of hornblende solid solution, much of the compositional area for mafic rocks lies in the two-phase Hbl-Plag field. Most amphibolites in hand specimen are thus predominantly black rocks with variable amounts of white plagioclase. An astute student might notice that the shaded field of common mafic compositions is close to hornblende, and that applying the lever principle suggests that the ratio of Hbl to Plg should range from 1.0 to about 0.7. Amphibolites may have more plagioclase than this because *ACF* diagrams, such as Figure 8, are projected from albite (a component of plagioclase). If the albite component is appreciable, the amount of plagioclase would be un-

position to “ocean-floor metamorphism,” in which the basalts of the oceanic crust were hydrothermally altered by hot recirculating brines in the vicinity of mid-ocean ridges (Vallance, 1967). Although cordierite is absent in Figure 8, it occurs in many amphibolite facies rocks, especially at lower pressure and higher temperature.

The transition from amphibolite to granulite facies occurs in the range 650 to 850°C. In the presence of an aqueous fluid, associated pelitic and quartzofeldspathic rocks (including granitoids) begin to melt in this range at low to medium pressures (see the solidus in Figure 2). Migmatites may form and the melts may become mobilized. As a result, not all pelites and quartzofeldspathic rocks reach the granulite facies prior to melting. H_2O is typically removed by the partial melts, however, and the remaining rocks may become depleted in H_2O , developing granulite facies mineral assemblages. Mafic rocks generally melt at somewhat higher temperature, and granulite facies mineral assemblages are common at high metamorphic grades. Hornblende, typically the last remaining major hydrous phase, decomposes and orthopyroxene + clinopyroxene appear. This reaction occurs over a temperature interval of at least 100°C.

The **granulite facies** is characterized by the presence of a largely anhydrous mineral assemblage. In metabasites the critical mineral assemblage is orthopyroxene + clinopyroxene + plagioclase + quartz. Garnet is also common, and minor hornblende and/or biotite may be present. In pelitic

Figure 25-8 from John D. Winter (p. 576): *ACF* compatibility diagram illustrating representative mineral assemblages in metabasites of the granulite facies. Typical mafic compositions fall squarely within the **Plagioclase–Clinopyroxene–Orthopyroxene** or **Plagioclase–Clinopyroxene–Garnet–Orthopyroxene** fields.

Key petrological features of granulite facies metabasites:

- **Coexistence of Two Pyroxenes:** The essential defining hallmark of granulite facies metabasites is the stable coexistence of *orthopyroxene* (*hypersthene/enstatite*) and *clinopyroxene* (*diopside/augite*) with plagioclase.
- **Anhydrous Character:** Hydrous minerals (hornblende, biotite, epidote) are absent or present only as minor retrograde/relict phases or F/Cl-enriched accessory crystals.
- **Plagioclase Composition:** Plagioclase is intermediate to calcic (andesine to labradorite/bytownite). At even higher pressures (transitioning to eclogite), plagioclase breaks down completely into omphacite + garnet + kyanite + quartz.



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 5

Question Statement

Q.5: Which one of the following is a characteristic assemblage of granulite facies metamorphism of mafic protolith?

(a) Hornblende + Plagioclase + Quartz + Garnet

(b) Forsterite + Orthopyroxene + Clinopyroxene + Epidote

(c) Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz

(d) Forsterite + Orthopyroxene + Clinopyroxene + Staurolite

Correct Answer: (c) Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz



The stable coexistence of two pyroxenes (Cpx + Opx) with plagioclase, garnet, and quartz represents the quintessential high-pressure/high-temperature anhydrous assemblage of mafic granulites.

Exhaustive Option-by-Option Critical Analysis

(c) Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz

CORRECT

Why it is uniquely correct: In metamorphic petrology (Winter, 2010, Table 25-1, p. 571; Yardley, 1989; Bucher & Grapes, 2011), the entry into the **granulite facies** in mafic rocks is universally defined by the dehydration of hornblende to produce **orthopyroxene** (hypersthene) in the presence of **clinopyroxene** and **plagioclase**. At medium to high pressures, almandine-pyrope **garnet** and **quartz** coexist stably with both pyroxenes. This anhydrous 5-phase assemblage (*Cpx + Opx + Plag + Grt + Qtz*) is the textbook definition of a mafic "two-pyroxene granulite."

(a) Hornblende + Plagioclase + Quartz + Garnet

INCORRECT

Why it is incorrect: This assemblage is the classic paragenesis of the **Amphibolite Facies** (specifically, garnet amphibolite). Hornblende is a hydrous amphibole whose upper thermal stability limit defines the boundary between amphibolite and granulite facies. When a mafic rock enters the granulite facies, hornblende breaks down completely by dehydration reactions to generate pyroxenes. Therefore, an assemblage dominated by hornblende cannot represent the granulite facies.

(b) Forsterite + Orthopyroxene + Clinopyroxene + Epidote

INCORRECT

Why it is incorrect: This option contains two severe petrological impossibilities for mafic granulites:

1. **Forsterite (Mg-olivine):** Mafic rocks (basalts/gabbros) are silica-saturated to oversaturated; olivine cannot coexist in equilibrium with quartz-bearing mafic granulites. Forsterite is restricted to silica-undersaturated *ultramafic* rocks (peridotites) or siliceous dolomitic marbles.
2. **Epidote:** Epidote is a hydrous Ca-Al sorosilicate characteristic of low-to-medium grades (greenschist and epidote-amphibolite facies, $T < 500^{\circ}\text{C}$ – 550°C). It completely decomposes at temperatures hundreds of degrees below the onset of granulite facies conditions.

(d) Forsterite + Orthopyroxene + Clinopyroxene + Staurolite

INCORRECT

Why it is incorrect: This assemblage is a chemically contradictory mixture:

1. **Staurolite:** Staurolite is strictly confined to highly aluminous, Fe-rich *pelitic* protoliths (metamorphosed shales) at amphibolite facies (520°C – 650°C). It never occurs in mafic bulk compositions because basalts lack sufficient Al_2O_3 relative to Ca, Mg, and Fe.
2. **Forsterite + Staurolite incompatibility:** Forsterite represents extreme silica-undersaturation and ultramafic compositions, whereas staurolite requires high Al and excess silica. They are chemically mutually exclusive in nature.



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Comprehensive Facies Paragenesis in Mafic Rocks & 8 Exam-Grade Multiple-Choice Questions

Comparative Metamorphic Facies Mineralogy of Mafic Protoliths (Basalts/Gabbros)

Metamorphic Facies	Diagnostic Mafic Assemblage	Key Stable Ca-Amphibole / Pyroxene	Plagioclase Status	P-T Regime
Zeolite	Laumontite/Heulandite + Chlorite + Albite + Quartz	None	Albite	T < 250°C, Low P
Prehnite-Pumpellyite	Prehnite + Pumpellyite + Chlorite + Albite + Quartz	None	Albite	250 – 350°C, Low-Mod P
Greenschist	Actinolite + Chlorite + Epidote + Albite + Quartz	Actinolite	Albite (<An ₅)	350 – 500°C, Mod P
Epidote-Amphibolite	Hornblende + Epidote + Albite/Oligoclase ± Garnet	Hornblende (aluminous)	Albite → Oligoclase	450 – 550°C, Mod P
Amphibolite	Hornblende + Plagioclase (An ₁₅ –An ₅₀) + Quartz ± Garnet	Hornblende	Oligoclase–Andesine	550 – 750°C, Mod-High P
Granulite	Clinopyroxene + Orthopyroxene + Plagioclase ± Garnet ± Quartz	Two Pyroxenes (Opx + Cpx)	Andesine–Bytownite	> 700 – 900°C, Mod-High P
Blueschist	Glaucophane + Lawsonite/ Epidote + Jadeite + Quartz	Glaucophane (Na-amphibole)	Plagioclase Unstable (Jadeite + Qtz)	200 – 450°C, High P (>6–8 kbar)
Eclogite	Omphacite + Pyrope-Almandine Garnet ± Kyanite ± Quartz	Omphacite (Na-Ca pyroxene)	Plagioclase Completely Absent	> 500°C, Very High P (>12–15 kbar)

Diagnostic Petrographic Criteria: Identifying Mafic Granulites

- **Presence of Orthopyroxene:** The definitive petrographic criterion separating granulite facies from amphibolite facies is the appearance of hypersthene/orthopyroxene (distinguished by faint pink-to-green pleochroism, parallel extinction, and low birefringence).
- **Two-Pyroxene Geothermometry:** Granulite metabasites preserve equilibrium compositions of coexisting clino- and orthopyroxenes that record peak metamorphic temperatures between 750°C and 950°C.
- **Granoblastic Polygonal Texture:** Mafic granulites typically display a granoblastic polygonal (or granoblastic mosaic) texture with straight grain boundaries meeting at ~120° triple junctions, indicating prolonged static annealing at high temperature.
- **Plagioclase Twinning:** Plagioclase exhibits well-developed albite and pericline twinning, frequently showing antiperthitic exsolution lamellae of K-feldspar.

8 High-Yield Practice Questions on Granulite Facies & Mafic Metamorphism

1. The critical mineral whose first appearance in metabasites defines the onset of the granulite facies is:

- (a) Almandine garnet (b) Orthopyroxene (hypersthene) (c) Diopside (d) Sillimanite

Answer: (b) Orthopyroxene (hypersthene) — In mafic rocks, the transition from amphibolite to granulite facies is definitively marked by the dehydration of hornblende to produce orthopyroxene in the presence of clinopyroxene and plagioclase.

2. Which reaction represents the dehydration breakdown of hornblende at the amphibolite-to-granulite facies boundary?

- (a) Hornblende + Quartz = Orthopyroxene + Clinopyroxene + Plagioclase + H₂O
(b) Actinolite + Epidote = Hornblende + Albite + H₂O
(c) Glaucophane + Quartz = Jadeite + Talc + H₂O
(d) Tremolite + Calcite + Quartz = Diopside + CO₂

Answer: (a) — Reaction (a) is the fundamental devolatilization reaction responsible for the classic two-pyroxene granulite assemblage.

3. Mafic granulites are often called "two-pyroxene granulites" because they contain:

- (a) Diopside and Augite
(b) Orthopyroxene (hypersthene/enstatite) and Clinopyroxene (augite/diopside)

- (c) Omphacite and Aegirine
- (d) Wollastonite and Hedenbergite

Answer: (b) — Two-pyroxene granulites are defined by the coexistence of an orthopyroxene (Ca-poor) and a clinopyroxene (Ca-rich).

4. Why is epidote completely absent from equilibrium granulite facies mafic rocks?

- (a) Epidote only forms in granite pegmatites
- (b) Epidote is a hydrous mineral that breaks down via dehydration at $T > 550^{\circ}\text{C}$ – 600°C
- (c) Epidote requires CO_2 to be stable
- (d) Epidote is converted directly to chlorite

Answer: (b) — Epidote is thermally limited to greenschist and epidote-amphibolite facies; it dehydrates to form anorthite-rich plagioclase, grossular, and hornblende well before reaching granulite temperatures.

5. At ultra-high temperatures (UHT metamorphism, $T > 900^{\circ}\text{C}$ – 1000°C) in pelitic granulites, which diagnostic anhydrous assemblage is recognized?

- (a) Sapphirine + Quartz
- (b) Chlorite + Muscovite
- (c) Glaucophane + Jadeite
- (d) Prehnite + Pumpellyite

Answer: (a) Sapphirine + Quartz — The coexistence of sapphirine + quartz or osumilite + quartz is the definitive indicator of ultra-high temperature (UHT) granulite metamorphism.

6. In mafic rocks, the transition from granulite facies to eclogite facies at high pressure is characterized by the disappearance of:

- (a) Garnet
- (b) Plagioclase
- (c) Clinopyroxene
- (d) Quartz

Answer: (b) Plagioclase — At high lithostatic pressures (> 12 – 15 kbar), plagioclase breaks down completely into jadeite/omphacite + grossular + kyanite + quartz, defining the eclogite facies.

7. Who first established the concept of metamorphic facies based on mineral assemblages in mafic rocks?

- (a) George Barrow
- (b) Pentti Eskola
- (c) N. L. Bowen
- (d) F. J. Turner

Answer: (b) Pentti Eskola — Eskola (1915, 1920) pioneered the metamorphic facies classification based on his study of Orijärvi and Oslo rocks.

8. What texture is most typical of mafic granulites that have undergone extensive static thermal annealing in the deep crust?

- (a) Nematoblastic
- (b) Slaty cleavage
- (c) Granoblastic polygonal
- (d) Poikilitic

Answer: (c) Granoblastic polygonal — High-temperature static recrystallization (annealing) drives grain-boundary migration to minimize interfacial surface energy, producing equant grains with $\sim 120^\circ$ dihedral angles.